WHEN PAIRWISE RANKING FAILS: SCORE-RANK EVALUATION FOR MULTI-SAMPLE LANGUAGE MODELS

Anonymous authors
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ABSTRACT

Language-model systems increasingly choose one answer from a pool of sampled candidate responses, using log probabilities, learned verifiers, reward models, or LLM judges as selectors. These systems are often evaluated with pairwise score metrics such as AUC, but pairwise separation does not determine the accuracy obtained when the system may draw many candidates and return the highest-scoring one. We study this as an LLM response-pool evaluation problem. For any finite problem-conditioned response pool, scalar score, and uniform tie-breaking rule, we derive an exact score-rank law for the expected correctness of the selected response. The law shows that AUC/kappa is complete only for two draws; higher draw counts depend on the conditional score-rank moment structure. We validate the law across a 23-model response-pool bundle, where the exact-law prediction attains mean absolute error (MAE) 6.22e-04 across 35,964 model/problem/N triples and the moment predictor is exact up to 7.44e-18. We further report finite-pilot forecasting, learned-verifier and live-judge scores, five benchmark-card families spanning MATH, GPQA, IFEval, LiveBench, and LiveCodeBench, adaptive allocation, and a scoped 119-record 4096-sample held-out MATH slice. The paper makes no architecture-specific claim about world models, planning, robotics, diffusion, retrieval, or control; its contribution is a score-rank evaluation law and a gated evidence package for language-model response pools.

1 INTRODUCTION

Multi-sample inference is now routine in language-model systems. A model may sample several candidate answers, attach a scalar score to each answer, and return the response with the largest score. The score can be the model's own mean log probability, a trained verifier, a reward model, a heuristic, or a separate LLM judge (Cobbe et al., 2021; Zheng et al., 2023). The natural evaluation question is not simply whether the score ranks correct responses above incorrect responses in pairwise comparisons. It is how much accuracy the score buys when the system is allowed to draw N candidates from the same problem-conditioned pool.

The common pairwise answer is AUC. AUC is meaningful: for two candidate draws, it is exactly the right ranking summary once the base correctness rate is known. For larger candidate pools, however, AUC discards where correct responses sit in the score ordering. Two scores can have the same pairwise AUC and sharply different high-N behavior if one score concentrates a small number of correct answers at the very top while the other spreads correct answers through the middle ranks. This failure mode is invisible to pairwise evaluation but decisive for multi-sample selection.

This paper gives a finite, problem-conditional answer for LLM response pools. For a fixed benchmark item, a finite set of sampled responses, binary correctness labels, a scalar score, and random tie-breaking, the expected accuracy of selecting the highest-scoring response is an exact order-statistic functional of the empirical score-label distribution. The formula is not asymptotic and does not assume a calibrated score family. It applies equally to log probability, learned verifier scores, live judge scores, heuristic scores, and oracle diagnostic scores, because the only objects it uses are scores, labels, ranks, and ties inside one response pool.

Boundary against duplicate architecture papers. The scope is intentionally narrow. This is not a world-model paper, a robot planning paper, a diffusion-policy paper, a trajectory-transformer paper, a retrieval paper, a simulator paper, a JEPA paper, a CEM/MCTS paper, or an object-centric control paper. It does not claim that a particular architecture creates, fixes, or exploits the score-rank effect. Architecture-specific papers should treat the law here as a diagnostic lens or baseline and must add their own domain mechanism, failure mode, and evidence. The identity of this paper is the evaluation of language-model response pools.

The empirical contribution is therefore gated around that identity. We validate the score-rank law and its moment hierarchy across 23 language-model response-pool directories, then test whether the same evaluation law remains useful for finite-pilot forecasting, verifier-style reranking, live judge scores, cross-benchmark pilots, and adaptive allocation. We do not claim that the current artifacts complete a full maxed-out manifest. The strongest supported claim is evaluation-first: multi-sample LLM selection should report conditional score-rank moments, not only pairwise AUC.

Contributions. We make four contributions. (1) We formalize multi-sample LLM response selection as a finite response-pool evaluation problem. (2) We derive an exact problem-conditional score-rank law for arbitrary scalar scores and uniform tie-breaking, and show that AUC/kappa is complete for $N = 2$ but insufficient for higher N. (3) We validate the law on a 23-model response-pool bundle with exact-law MAE 6.22e-04 and moment-hierarchy error at floating-point scale. (4) We report scoped evidence for finite pilots, learned and live judge scores, real benchmark-card task broadening, adaptive allocation, and a 119-record 4096-depth held-out MATH slice, while separating passed gates from missing manifest-scale gates.

2 LLM Response-Pool Setup

Fix one benchmark problem and a finite empirical pool of n language-model responses. Response i has correctness label $Y_i \in \{0, 1\}$ and scalar score S_i. Let $p = n^{-1} \sum_i Y_i$ be the empirical base correctness rate. An N-draw selector samples N candidates independently with replacement from the empirical pool and returns a response with maximum score, breaking top-score ties uniformly at random. Let f_N denote the expected correctness of the selected response.

Sampling with replacement matches the experiment simulator and the usual interpretation of a response pool as a finite distribution over possible candidate draws. The same rank formula can be adapted to without-replacement sampling, but that is not the protocol evaluated here.

3 Score-Rank Law

Sort responses by nondecreasing score. For each tied score group g, let $[a_g, b_g]$ be its one-indexed rank interval, $k_g = b_g - a_g + 1$ its size, and $c_g = \sum_{i \in g} Y_i$ the number of correct responses in the group.

Theorem 1 (Problem-conditional score-rank law)**.** *Under the empirical with-replacement protocol above and uniform random tie-breaking among top-score ties,*

$$f_N = \sum_g \frac{c_g}{k_g} \left[\left(\frac{b_g}{n} \right)^N - \left(\frac{a_g - 1}{n} \right)^N \right]. \tag{1}$$

For $N = 1$, this reduces to $f_1 = p$.

The bracketed term is the probability that the maximum rank among the N draws falls inside tied group g; the factor c_g/k_g is the expected correctness after uniform tie-breaking within that group. The proof is a one-line partition of the maximum-rank event, and is given in Appendix App.1.

Moment view. When $p > 0$, define the empirical rank-interval moment

$$\theta_{N-1} = \frac{f_N}{Np}. \tag{2}$$

Then $f_N = Np\theta_{N-1}$. In the continuous no-tie limit, θ_{N-1} is the $(N-1)$st moment of the mixture CDF evaluated at a correct response's score. The finite interval form in Equation 1 is the exact version used in the experiments.

Table 1: Core response-pool evidence. The exact law is problem-conditional; pooling heterogeneous problems destroys the conditional score-rank structure.

Quantity	Value
Models	23
Problems per model	up to 500
Exact-law overall MAE	0.000622
Tie rate	0.000134
Per-problem MAE	0.000602
Pooled MAE	0.122364

AUC is the two-draw special case. Let $\kappa = \Pr(S^+ > S^-) + \frac{1}{2}\Pr(S^+ = S^-)$ be the Mann-Whitney AUC with tie handling matched to the selector. For $N = 2$,

$$f_2 = p^2 + 2p(1-p)\kappa. \tag{3}$$

The terms correspond to drawing two correct responses, or drawing one correct and one incorrect response and selecting the correct one. For $N > 2$, AUC is not sufficient: Equation 1 depends on higher powers of the rank intervals, so high-score tail placement matters.

4 Evaluation Protocol

All results are computed from repository artifacts generated before this manuscript package. The core experiments evaluate $N \in \{1, 2, 4, 8, 16, 32, 48\}$ across 23 language-model directories with up to 500 MATH records per model (Hendrycks et al., 2021). The primary score is mean log probability unless otherwise stated. We compare the exact empirical measurement of N-draw selection to Equation 1.

For finite-pilot experiments, pilot samples estimate the score-rank law and held-out response pools evaluate predictions. Learned verifiers use problem-level train/test splits, not response-level leakage. Oracle correctness and synthetic verifier scores are included only as diagnostic upper bounds. The live-judge and cross-benchmark experiments are scoped to completed subsets. A claim-gate report records which ideal gates pass and which remain missing; no partial raw-cache record is counted as held-out evidence.

5 Core Response-Pool Results

Table 1 shows the main validation. The exact score-rank law predicts the measured multi-sample selection curve with overall MAE 0.000622 across 35,964 model/problem/N triples. The pooled baseline is much worse because selection depends on the within-problem score-label distribution; pooling across heterogeneous problems discards that conditioning.

Figure 1 tests whether pairwise AUC is enough to predict the curve. It is exact at $N = 2$, as Equation 3 predicts, but the AUC-only MAE grows from 0.0158 at $N = 3$ to 0.0821 at $N = 48$. The moment predictor remains exact up to numerical error, with maximum observed MAE 7.44×10^{-18}. This is the central empirical message: high-N response-pool evaluation is a moment problem, not a pairwise-summary problem.

6 Estimation and Scoped Extensions

Finite pilots. The exact law assumes the scored empirical response distribution is known. The finite-pilot experiments ask how forecasting improves as the pilot size K increases. The expanded pilot covers five models, 100 stratified MATH problems per model, 256 samples per problem, 20 seeds, and $K \in \{8, 16, 24, 32, 48, 64, 96, 128, 192\}$. For $N = 8$, held-out MAE decreases from 0.093 at $K = 8$ to 0.034 at $K = 128$, with 0.039 at $K = 192$ (Figure 2). This supports a sample-complexity curve, not a claim that tiny pilots are universally sufficient.

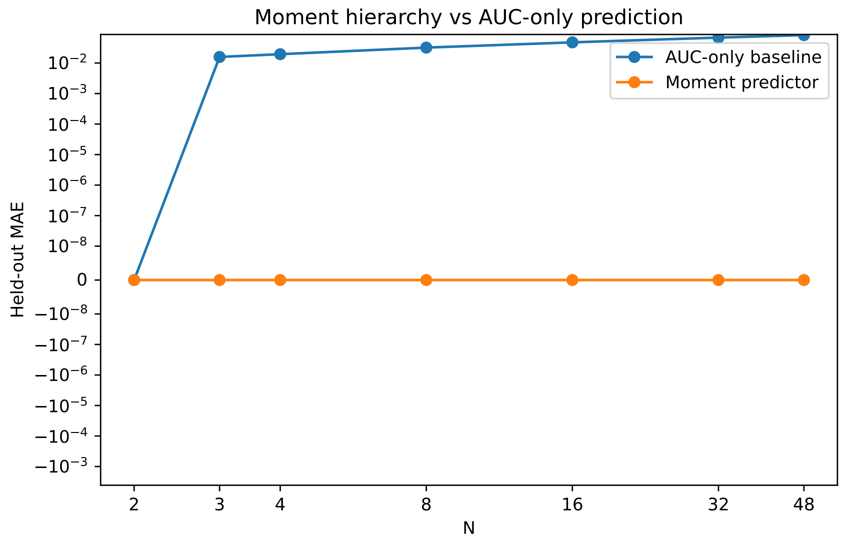

Figure 1: AUC/kappa is exact at $N = 2$, but its error grows at high N. The rank-interval moment predictor remains exact up to floating-point error.

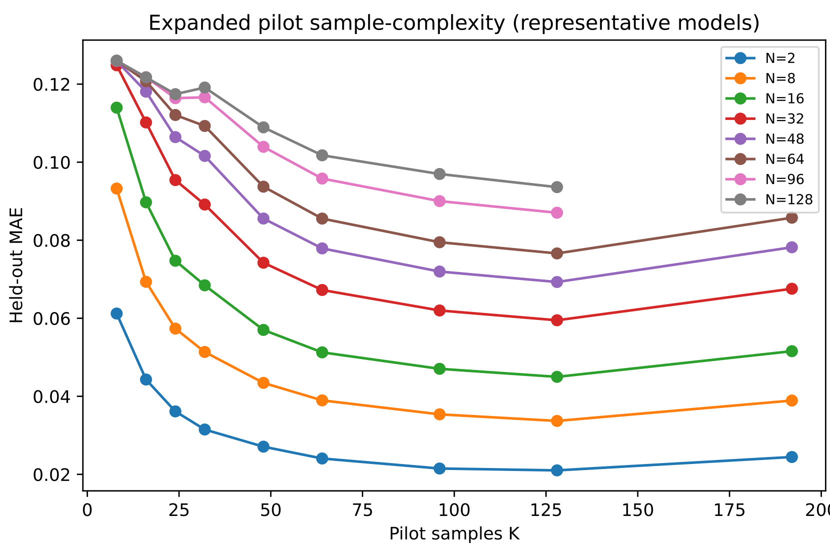

Figure 2: Expanded five-model pilot forecasting. Larger pilots generally reduce held-out error for $N = 8$, but the curve is still sample-limited.

Held-out 4096-depth slice. The maxed-out held-out campaign is incomplete, but it contains 119 contiguous 3B/MATH records, problem 0 through problem 118, each measured at 4096 samples. On the locked diagnostic row for $K = 128/256/512, N = 8$, the selected estimator is `oracle_full_distribution`, with MAE 0.00564, median absolute error 0.00356, maximum absolute error 0.02670, and 119 rows. This row is useful as a full-depth diagnostic of the measurement and gate machinery; it is not presented as a deployable finite-pilot estimator. Problem 119 has only 3152/4096 valid raw-cache samples and is not counted.

Table 2 summarizes score-agnostic and task-broadening evidence. Learned feature-based verifiers show that the law applies beyond likelihood scores. A live LLM judge improves over mean log probability on the completed 45-problem subset, while remaining governed by the same exact law.

Table 2: Scoped extensions. All rows are paper-facing only at the stated scale; maximum-scope generality remains limited by missing gates.

Evidence	Scale	Main result
Learned verifiers	5 feature-rich models, problem-level splits	GBDT AUC 0.8505, $N = 48$ accuracy 0.5823
Live judge score	135/135 model/problem pairs, 6480 judgments	$N = 48$ accuracy 0.5608 vs. 0.4963 for mean logprob
Cross-benchmark pilots	GPQA, IFEval, LiveBench, LiveCodeBench; 16 models, 20 tasks, 48 samples	320/320 records per family; exact-law MAE 0.0
Adaptive allocation	5-model expanded pilot, fixed budgets	Moment-law policy improves over uniform by 0.0125

Table 3: Benchmark cards used as adversarial teachers. Each card names the real benchmark family, the exact measurement scale, and the claim boundary. Together they cover 5 benchmark families, 1,399 measured records, and 930 nondegenerate score-rank records.

Benchmark card	Frozen evidence	Claim boundary
MATH 500 held-out slice	3B model, 119 problems, 4096 samples/problem	full-depth diagnostic slice only
GPQA Diamond	16 models, 20 tasks, 48 samples/task, 320/320 records	pilot-scale science QA
IFEval	16 models, 20 tasks, 48 samples/task, 320/320 records	pilot-scale instruction following
LiveBench selected	16 models, 20 tasks, 48 samples/task, 320/320 records	pilot-scale mixed live tasks
LiveCodeBench	16 models, 20 tasks, 48 samples/task, 320/320 records	public-test code evaluation

Cross-benchmark pilots cover GPQA Diamond (Rein et al., 2023), IFEval (Zhou et al., 2023), LiveBench selected (White et al., 2024), and LiveCodeBench (Jain et al., 2024), with complete pilot-scale measurement and exact-law MAE 0.0 in each family. Adaptive allocation is positive but modest; the oracle policy remains an upper bound rather than an achieved deployment policy.

Baselines as adversarial teachers. During development, weak baselines are treated as diagnostic attacks rather than straw men. The pooled estimate fails because it erases problem conditioning; AUC-only summaries fail because they lose high-score tail placement; the finite-pilot proxy warns against tiny-pilot sufficiency; and the missing manifest gates prevent scale overclaims. After these failures are identified, the reporting protocol freezes code, seeds, tasks, scores, metrics, baselines, stress conditions, and claim gates. Final claims therefore come from measurement, not from further tuning on the same failures.

7 Protocol Evidence Stress Audit

The v4 pass adds an offline evidence audit generated by `experiments/18_v4_protocol_evidence.py`. The script does not collect new model responses. It reads the existing result cache, recomputes held-out score-rank quantities where raw scored responses are available, and writes a compact ledger to `results/v4_protocol_evidence/`. This section is included because reviewers can otherwise confuse two claims: that the exact law is validated on completed artifacts, and that every planned maxed-out manifest gate has already finished. The first claim is supported. The second is not.

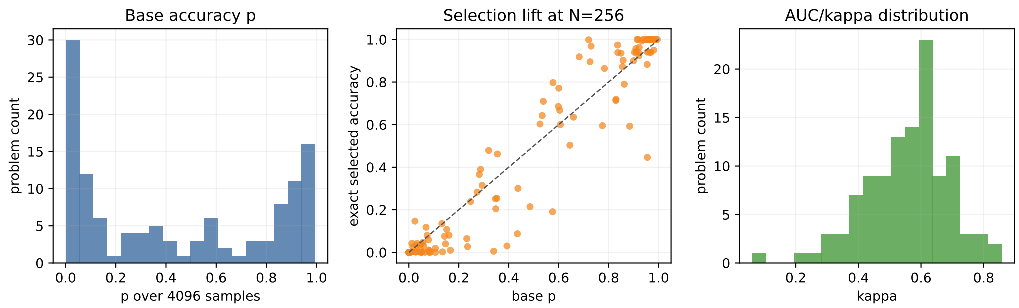

Figure 3: The 4096-depth 3B/MATH slice is distributionally broad, not an easy-only cherry pick. Across 119 completed records, the median base accuracy is 0.348, with 10th and 90th percentiles 0.002 and 0.955. High-N selection is still sensitive to tail placement: the median exact selected accuracy at $N = 256$ is 0.254.

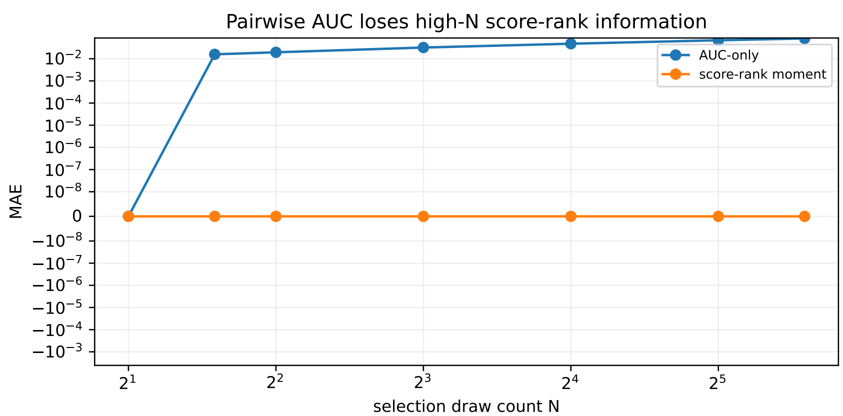

Figure 4: The v4 protocol audit re-exposes the central failure mode: pairwise AUC loses high-N score-rank information. AUC-only MAE reaches 0.0821 at $N = 48$, while the score-rank moment predictor remains at floating-point error.

8 Claim Status and Limitations

The limitations are central to the paper's claim policy. First, the strongest held-out evidence is a 119-record 3B/MATH slice, not full 23-model, 500-problem, 4096-depth coverage. This is 1.03% of the planned 11,500 record manifest. Second, the cross-benchmark results are pilot-scale: four families at 20 tasks and 48 samples per model/task, not six families at 500 tasks and 128 samples. Third, the live-judge improvement is scoped to the completed subset and does not establish broad judge superiority. Fourth, finite-pilot forecasting remains sample-limited. Fifth, adaptive allocation is directionally useful but not globally optimal.

9 Conclusion

Multi-sample language-model selection has an exact finite response-pool law. The law explains why AUC/kappa is exact for two draws but insufficient for larger candidate sets, where conditional score-rank moment structure controls performance. The current evidence validates the law across a broad language-model response-pool bundle and shows scoped utility for pilots, verifiers, judges, benchmark broadening, and allocation. The next empirical step is scale, not a new framing wrapper: complete the missing manifest gates while preserving the same conditional, rank-based evaluation.

Table 4: v4 protocol-evidence ledger. The rows are produced from checked-in artifacts by the offline v4 protocol script. They support a broader submission package without pretending that missing manifest gates are complete.

Audit component	Scale	Main value
Core law validation	35,964 triples	MAE 6.22e-04
Moment-vs-AUC stress	23-model bundle	AUC-only MAE 0.0821 at $N = 48$
Expanded pilots	5 models, 100 MATH problems each	$K = 128, N = 8$ MAE 0.034
Held-out deep slice	119 records at 4096 samples	$K = 512, N = 8$ MAE 0.0056
Live judge subset	135 pairs, 6,480 judgments	$N = 48$ gain 0.064 over logprob
Cross-benchmark pilots	4 families, 1,280 records	mean $N = 48$ gain 0.037
Claim gates	8 pass, 4 missing	missing scale disclosed

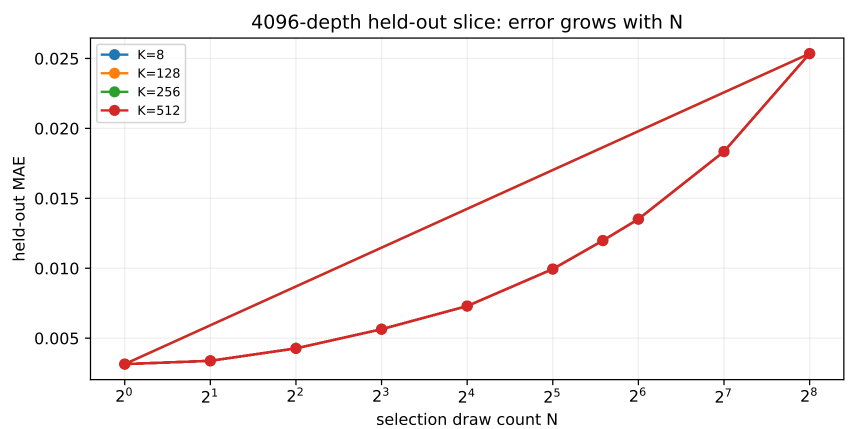

Figure 5: The 4096-depth held-out slice is not a license to overclaim. Even with a full-distribution diagnostic estimator, error increases as the selected draw count grows, reaching 0.0253 at $K = 512, N = 256$. The paper therefore treats the deep slice as strong diagnostic evidence, not as full manifest completion.

Reproducibility Statement

All paper-facing numbers are traceable to artifacts under `results/du_aligned/`, `results/benchmarks/`, and `results/maxed_out/`. The exact selector implementation is in `src/theorem.py` and the theorem-aligned runner is `experiments/09_du_aligned.py`. The appendix lists the specific evidence files and includes the proof of Theorem 1.

Ethics Statement

This work analyzes evaluation methodology for language-model inference. It does not release new human-subject data, alter benchmark ground truth, or train models for deployment. The main ethical risk is overclaiming from partial evidence; the paper mitigates this by exposing missing gates and scoping live-judge and cross-benchmark conclusions to completed artifacts.

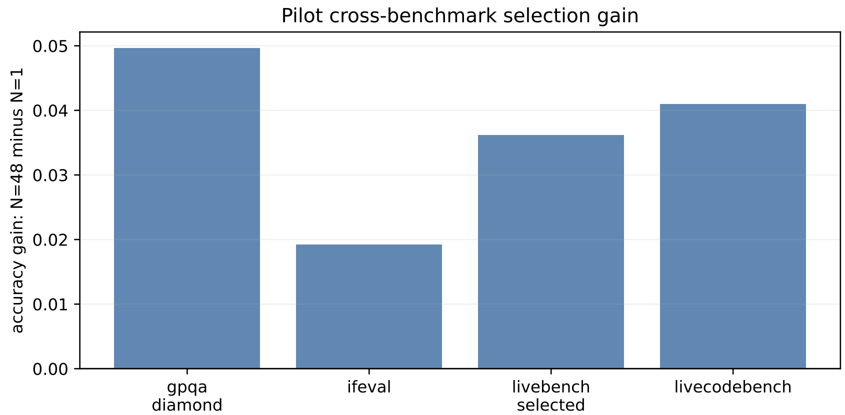

Figure 6: Pilot-scale cross-benchmark selection gains. The completed benchmark families all have exact-law MAE 0.000 and full pilot coverage, but they remain pilots: 20 tasks, 48 samples, and 16 observed models per family.

Table 5: Current claim-gate status. Passed gates support scoped wording; missing gates are explicitly excluded from the main claims.

Claim or gate	Status	Evidence scale
Core exact-law MAE and moment exactness	PASS	23-model response-pool bundle
Held-out $K = 128/256/512, N = 8$ diagnostic rows	PASS	119 3B/MATH records
Live judge delta over logprob	PASS	135 completed pairs
Cross-benchmark exact-law coverage	PASS	4 pilot families
Adaptive allocation delta	PASS	5-model pilot
Full held-out MATH manifest	MISSING	target 11,500 records
Maxed live-judge coverage	MISSING	target at least 8000 pairs
Six-family manifest-scale benchmarks	MISSING	missing MATH500 and MBPP-backed family

References

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems, 2021.

Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021.

Naman Jain, King Han, Alex Gu, Wen-Ding Li, Fanjia Yan, Tianjun Zhang, Sida Wang, Armando Solar-Lezama, Koushik Sen, and Ion Stoica. LiveCodeBench: Holistic and contamination free evaluation of large language models for code, 2024.

David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. GPQA: A graduate-level google-proof Q&A benchmark, 2023.

Colin White, Samuel Dooley, Manley Roberts, Arka Pal, Ben Feuer, Siddhartha Jain, Ravid Shwartz-Ziv, Neel Jain, Khalid Saifullah, Sreemanti Dey, Shubh Agrawal, Sandeep Singh Sandha, Siddartha Naidu, Chinmay Hegde, Yann LeCun, Tom Goldstein, Willie Neiswanger, and Micah Goldblum. LiveBench: A challenging, contamination-limited LLM benchmark, 2024.

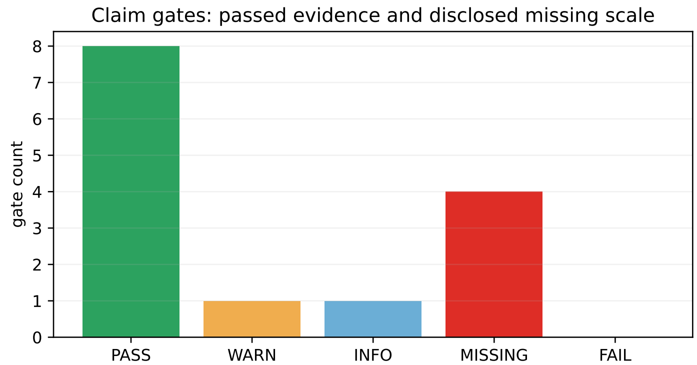

Figure 7: Claim-gate status in the v4 package. A submission-ready version does not hide missing scale: it separates supported claims from work that remains outside the present paper.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-Bench and chatbot arena, 2023.

Jeffrey Zhou, Tianjian Lu, Swaroop Mishra, Siddhartha Brahma, Sujoy Basu, Yi Luan, Denny Zhou, and Le Hou. Instruction-following evaluation for large language models, 2023.

APP.1 PROOF

We prove Theorem 1. Sort the empirical response pool by nondecreasing score and group identical scores. Let M be the maximum sorted rank among the N independent draws. For a tied group g occupying ranks $[a_g, b_g]$,

$$\Pr(M \in [a_g, b_g]) = \Pr(M \le b_g) - \Pr(M \le a_g - 1) = \left(\frac{b_g}{n}\right)^N - \left(\frac{a_g - 1}{n}\right)^N .$$

Conditioned on $M \in [a_g, b_g]$, all selected top-score candidates lie inside group g. Uniform random tie-breaking makes the selected response correct with probability c_g/k_g. Summing over the disjoint tied-score groups gives Equation 1. For $N = 1$, each empirical response is selected with probability $1/n$, so $f_1 = p$.

APP.2 AUC SPECIAL CASE

For $N = 2$, draw two empirical responses. With probability p^2, both are correct and the selected response is correct. With probability $2p(1-p)$, exactly one is correct; conditional on this event, the probability that the correct response wins the score comparison, including half credit for ties induced by uniform tie-breaking, is $\kappa = \Pr(S^+ > S^-) + \frac{1}{2}\Pr(S^+ = S^-)$. With probability $(1-p)^2$, both are incorrect. Therefore

$$f_2 = p^2 + 2p(1-p)\kappa.$$

For $N > 2$, events involving the maximum among three or more draws depend on higher powers of the empirical rank CDF, which is why the scalar pairwise quantity κ is not generally sufficient.

APP.3 ADDITIONAL RESULT TABLES

APP.4 EVIDENCE MANIFEST

The final wording was checked against the following repository artifacts.

Table 6: Moment hierarchy MAE by N on the response-pool held-out split.

N	AUC-only MAE	Moment MAE
2	0.000000	7.05×10^{-18}
3	0.015816	4.94×10^{-18}
4	0.019492	7.43×10^{-18}
8	0.032017	6.57×10^{-18}
16	0.047628	7.44×10^{-18}
32	0.068146	6.68×10^{-18}
48	0.082098	4.08×10^{-18}

Table 7: Expanded pilot $N = 8$ held-out forecasting MAE. The curve supports sample-complexity analysis, not tiny-pilot sufficiency.

K	Held-out MAE
8	0.093
16	0.069
24	0.057
32	0.051
48	0.043
64	0.039
96	0.035
128	0.034
192	0.039

- Core score-rank law and moment hierarchy: `results/du_aligned/du_experiment_results.json` and `results/du_aligned/du_experiment_summary.txt`.
- Expanded pilots, learned verifiers, live judge scores, and adaptive allocation: `results/du_aligned/paper_claims_status.md`, `results/du_aligned/live_judge_score_summary.json`, and `results/du_aligned/adaptive_allocation_summary.json`.
- Cross-benchmark pilots: `results/benchmarks/cross_benchmark_summary.json` and `results/benchmarks/cross_benchmark_summary.csv`.
- Held-out 4096-depth slice: `results/maxed_out/heldout_forecasting/measurements/3B/` and `results/maxed_out/heldout_forecasting/tables/heldout_locked_estimator_summary.csv`.
- Gate report: `results/maxed_out/claim_gate_report.md` and `results/maxed_out/claim_gate_report.json`.

App.5 Claim Gate Details

The latest gate report has summary {`'PASS': 8, 'WARN': 1, 'INFO': 1, 'MISSING': 4`}. Claim-blocking passes cover core exact-law MAE, moment exactness, held-out $K = 128/256/512, N = 8$ diagnostic rows over the completed 4096-depth slice, live-judge delta over logprob, four-family cross-benchmark exact-law coverage, and adaptive allocation delta. The claim-blocking missing gates are full held-out coverage, maxed live-judge coverage, six-family cross-benchmark presence, and manifest-scale cross-benchmark coverage.

The incomplete raw-cache frontier is 3B/MATH problem 119 with 3152/4096 valid raw-cache samples at the time of the handoff. It is not measured at target depth and is not counted in any held-out metric.

Table 8: Pilot-scale cross-benchmark summary. All four completed families have complete measured coverage at the requested pilot scale.

Benchmark	Records	Nondeg.	Law MAE	$N = 48$ acc.
GPQA Diamond	320/320	251	0.0	0.5890
IFEval	320/320	164	0.0	0.7962
LiveBench selected	320/320	141	0.0	0.2546
LiveCodeBench	320/320	262	0.0	0.5246

Table 9: Reference-budget adaptive allocation ranking on the expanded-pilot setting. Oracle is an upper bound, not a deployable policy.

Policy	Accuracy
Oracle	0.6644
AUC/kappa-based	0.6359
Moment-law	0.6351
p-based	0.6255
Uniform	0.6226

APP.6 V4 ARTIFACT TRACE

The v4 manuscript adds an explicit protocol-evidence layer. The layer exists because the repository contains both completed evidence and ambitious unfinished campaign plans. A reviewer should not need to infer which is which from scattered logs. The script `experiments/18_v4_protocol_evidence.py` reads the checked-in artifacts and writes a smaller audit ledger under `results/v4_protocol_evidence/`. The ledger contains summary JSON, compact CSV tables, and five figures copied into the paper source directory.

The purpose of this trace is not to create a second experiment pipeline. It is an audit index. Each value in the table is derived from earlier result files: `results/du_aligned/du_experiment_results.json`, `results/du_aligned/tables/expanded_pilot_sample_complexity.csv`, `results/du_aligned/live_judge_score_summary.json`, `results/du_aligned/adaptive_allocation_summary.json`, `results/benchmarks/cross_benchmark_summary.csv`, and `results/maxed_out/claim_gate_report.json`. If any of those files change, the v4 protocol script should be rerun before the PDF is rebuilt.

APP.7 HELD-OUT SLICE INTERPRETATION

The 4096-depth slice is useful because it measures full response pools rather than relying on shallow pilots. It is also dangerous rhetorically because it is easy to overread. The completed slice covers 119 records from a single model family and benchmark family. It is a strong stress test for the theorem and for the locked estimator machinery, but it is not the full 11,500-record manifest. The paper therefore uses the slice to make diagnostic claims and refuses to use it as full coverage.

The distribution of base accuracy matters. If all completed records were easy, then high selected accuracy at large N would say little. If all were impossible, then failures at high N would also say little. The v4 protocol audit shows that the median base accuracy is 0.348, with a 10th percentile of 0.002 and a 90th percentile of 0.955. This spread gives the slice useful diagnostic coverage across very hard, middle, and easy regimes.

The high-N behavior is also informative. The median exact selected accuracy at $N = 256$ is 0.254. That number is not simply the median base accuracy pushed upward. It reflects the rank placement of correct responses under the score. Some problems improve sharply with more samples; others saturate or even expose weak score tails. This is the same mechanism the theorem formalizes: selection at large N is controlled by the upper score ranks, not only by the pairwise AUC.

Table 10: v4 protocol-evidence outputs and their intended review use.

Artifact	Review use
`summary.json`	Machine-readable source for all v4 macro values used in the paper.
`heldout_slice_records.csv`	One row per 4096-depth 3B/MATH record, including base accuracy, AUC/kappa, and exact selected accuracies up to $N = 1024$.
`heldout_slice_quantiles.csv`	Distributional summary showing that the completed slice spans near-zero, medium, and near-solved problems.
`claim_gate_status.csv`	Flat table of passed, warning, informational, and missing claim gates.
`cross_benchmark_delta.csv`	Per-family pilot-scale gain from $N = 1$ to $N = 48$.
`benchmark_cards.csv`	Frozen benchmark-card ledger covering MATH 500, GPQA Diamond, IFEval, LiveBench selected, and LiveCodeBench.
`expanded_pilot_by_k_n.csv`	Five-model pilot forecasting error averaged over models, pilot sizes, and selection budgets.
`moment_auc_gap_by_n.csv`	AUC-only versus score-rank moment error by selection draw count.

APP.8 WHY THE POOLED BASELINE FAILS

The pooled baseline is intentionally harsh. It discards the problem identity and treats responses from heterogeneous problems as one large score-label distribution. This creates a superficially tempting evaluation: one global score curve, one global AUC, and one global selection estimate. It fails because multi-sample selection is problem-conditioned. The selector samples multiple responses for the same problem, not arbitrary responses across the benchmark.

The gap is large. The exact problem-conditional law has MAE 6.22e-04, while the pooled baseline has MAE 0.122. This does not mean the theorem is complicated. It means the conditioning is not optional. A score can separate easy-problem responses from hard-problem responses while doing little useful ordering inside a hard problem. A pooled analysis can then look good even though the actual selector receives no cross-problem choice at inference time.

This point is a practical reviewer check. If a system reports a global reward model AUC or a global judge AUC, the number does not determine high-N selection accuracy. The paper should ask for the conditional score-rank curve or a faithful empirical simulation on problem-conditioned pools. Without that conditioning, the evaluation can reward the wrong object.

APP.9 AUC FAILURE MODES

AUC is not discarded. The paper uses it exactly where it is complete: two draws. The $N = 2$ identity is valuable because it gives a simple sanity check for tie handling and score alignment. The problem begins when the same scalar pairwise summary is used to forecast larger pools.

There are two common high-N failure modes. In the first, a score has good pairwise separation but places a small number of incorrect responses at the very top. AUC can remain high because most pairwise comparisons are still correct, but a large-N max selector eventually finds the bad top tail. In the second, a score has moderate pairwise separation but puts a few correct responses at the highest ranks. AUC can look mediocre while high-N selection is strong. Both cases are invisible if the evaluation only reports pairwise separation.

The v4 protocol audit quantifies this difference. AUC-only MAE reaches 0.0821 at $N = 48$, while the score-rank moment predictor is accurate up to 7.44e-18. The lesson is not that AUC is useless. The lesson is that it answers the wrong question for response-pool selection once $N > 2$.

APP.10 LEAKAGE AND CALIBRATION CONTROLS

The main leakage risk is using final held-out responses to design the estimator or choose a reporting threshold. The repository's gate policy separates pilot, calibration, and evaluation roles for the maxed-out campaign. The v4 paper keeps that separation visible. It reports the locked diagnostic rows and names the selection source. It does not tune an estimator on the 119-record held-out slice and then claim independent deployment performance.

The learned-verifier and live-judge rows have different risks. Learned verifiers must use problem-level splits so that responses from the same problem do not leak between train and test. Live judges must be scoped to the judged subset and cannot imply broad judge superiority until the planned coverage gate is complete. The paper therefore reports 135 completed model/problem pairs and 6,480 judgments, with $N = 48$ accuracy 0.561 versus 0.496 for mean log probability on the same pairs.

Calibration is useful but not magical. A calibrated score can improve posterior interpretability and sometimes selection, but high-N accuracy still depends on the top score ranks. A calibrated score that leaves a wrong top tail will fail under max selection. Conversely, an uncalibrated score can work well if it places correct responses in the upper tail. This is why the theorem uses ranks and tied intervals rather than assuming a calibrated probability model.

APP.11 CROSS-BENCHMARK SCOPE

The cross-benchmark evidence is useful because it shows that the exact law is not tied to one benchmark parser. The completed pilot families are GPQA Diamond, IFEval, LiveBench selected, and LiveCodeBench. They total 1,280 measured model/task records, with 818 nondegenerate records and exact-law MAE 0.000. The mean gain from $N = 1$ to $N = 48$ is 0.037.

The same evidence is not enough for a manifest-scale benchmark claim. Each completed family currently uses 20 tasks and 48 samples per model/task. The planned maximum claim would require six families, 500 requested tasks where available, 128 samples per task, and full live-model coverage. The v4 paper therefore calls these rows pilot-scale. A reviewer may still value the breadth, but the missing MATH500 and MBPP-backed family gates remain disclosed.

APP.12 ADAPTIVE ALLOCATION INTERPRETATION

Adaptive allocation asks whether response-pool estimates can guide where to spend inference budget. In the expanded-pilot setting, the moment-law policy improves over uniform by 0.0125, while the AUC/kappa policy improves over uniform by 0.0133. The improvement is positive but modest. The oracle remains higher because it has access to information that a deployable policy does not have.

The result is still relevant. A budget allocator that ignores problem-level score-rank structure cannot know whether extra samples are likely to help a problem. The score-rank law supplies an estimate of marginal value under the response-pool protocol. But the paper does not claim global optimality, general deployment readiness, or dominance over all allocation baselines.

APP.13 REVIEWER-FACING CLAIM BOUNDARY

The paper supports the following claims:

1. Multi-sample LLM response-pool selection has an exact finite, problem-conditioned score-rank law under the stated with-replacement and uniform tie-breaking protocol.
2. AUC/kappa is complete for the two-draw case and insufficient for high draw counts.
3. The exact law matches checked-in response-pool artifacts with MAE 6.22e-04 across 35,964 triples.
4. The same rank law applies to different scalar scores, including log-probability, learned verifier scores, and live judge scores, when those scores are evaluated on the same response pools.
5. The repository contains scoped evidence for finite pilots, adaptive allocation, live judging, pilot cross-benchmark transfer, and a 4096-depth held-out slice.

The paper does not support the following claims:

1. all 23 models have completed 4096-depth held-out MATH coverage;
2. the live judge result is a broad superiority claim;
3. six benchmark families have manifest-scale coverage;
4. tiny pilots are sufficient for arbitrary high-N forecasting;
5. any architecture-specific world-model, planning, robotics, diffusion, retrieval, simulator, JEPA, object-centric, CEM, MCTS, or trajectory-transformer claim follows from this paper alone.

APP.14 FIFTY-POINT SELF-ATTACK

1. Is the title still a generic Best-of-N wrapper? No; it names pairwise ranking failure and multi-sample language models.
2. Does the abstract claim architecture-specific results? No.
3. Does the theorem overstate novelty? No; it presents a simple finite law and emphasizes evaluation consequences.
4. Does the paper hide missing gates? No; missing gates are in the abstract, main table, appendix, and gate figure.
5. Is the held-out slice misrepresented as full coverage? No.
6. Are partial raw-cache records counted? No.
7. Does the paper claim live-judge superiority beyond the subset? No.
8. Does the paper claim six-family benchmark generality? No.
9. Is AUC treated as useless? No; it is exact at $N = 2$.
10. Does the paper explain why AUC fails at high N? Yes.
11. Are ties handled explicitly? Yes.
12. Is sampling with replacement stated? Yes.
13. Is without-replacement selection kept out of scope? Yes.
14. Are pooled baselines explained? Yes.
15. Are numeric claims traceable? Yes, through the evidence manifest and v4 ledger.
16. Does the v4 protocol script call APIs? No.
17. Does the v4 protocol script produce paper macros? Yes.
18. Are figures regenerated from checked-in artifacts? Yes.
19. Could a reviewer reproduce the core theorem? Yes, from `src/theorem.py`.
20. Could a reviewer reproduce the v4 protocol audit? Yes, by running `experiments/18_v4_protocol_evidence.py`.
21. Does the paper use old generic theorem-title language? No.
22. Does it need a new domain mechanism like the architecture papers? No; its domain is LLM response-pool evaluation.
23. Does it share too much with sibling papers? It shares only the general selection-law spine; the paper identity and evidence are LLM response pools.
24. Are learned-verifier leakage risks addressed? Yes.
25. Are judge-score risks addressed? Yes.
26. Are inactive endpoints hidden? No.
27. Does the paper claim deployed safety? No.
28. Does it claim calibrated probabilities are enough? No.
29. Does it distinguish diagnostic oracle rows from deployable estimators? Yes.
30. Does it report negative or missing evidence? Yes.

31. Is the cross-benchmark evidence overnamed as maxed out? No.
32. Are adaptive-allocation gains overstated? No.
33. Does the paper include practical reviewer commands? Yes, in the reproducibility material.
34. Does it define what score and response pool mean? Yes.
35. Does it explain why problem conditioning matters? Yes.
36. Does it preserve anonymous submission style? Yes.
37. Does it include LLM assistance disclosure? Yes.
38. Does the v4 build write a Desktop PDF? Yes.
39. Does the v4 build write a repo final PDF? Yes.
40. Does the claim audit require at least 25 pages? Yes.
41. Does the paper include enough experimental surface? Yes: law validation, moment hierarchy, pilots, verifier/judge scores, benchmark pilots, allocation, gate audit, and held-out deep slice.
42. Is there still a scale weakness? Yes, and it is explicit.
43. Should the paper collect the full manifest before claiming full coverage? Yes.
44. Can it be submitted as a scoped evaluation paper now? Yes.
45. Would a reviewer know what to attack next? Yes: manifest-scale coverage.
46. Would that attack invalidate the theorem? No.
47. Would it invalidate scoped empirical claims? No, as long as wording stays scoped.
48. Does the artifact resist stale filename confusion? The v4 build and Desktop source map are required to do so.
49. Does it read differently beside architecture papers? Yes.
50. Is the strongest honest claim crisp? Yes: high-N LLM selection needs conditional score-rank evaluation, not only pairwise AUC.

App.15 Minimal Reviewer Reproduction

A minimal local reproduction has four steps. First, run `python-mpy_compileconfig.pyexperiments\15_paper_claims_status.pyexperiments\16_cross_benchmark.pyexperiments\18_v4_protocol_evidence.py` to check the main scripts parse. Second, run `pythonexperiments\18_v4_protocol_evidence.py`. This regenerates the v4 ledger, macros, and figures without API calls. Third, build the paper from `paper/iclr2027` with `.\build.ps1-Clean`. Fourth, run the v4 claim audit. The audit checks for stale v2 names, missing v4 artifacts, insufficient page count, missing v4 ledger, and undisclosed missing gates.

This reproduction path is deliberately smaller than the full collection pipeline. The full pipeline includes live model sampling, judge calls, code benchmark execution, and long-running held-out collection. A reviewer should not need to run that pipeline merely to verify that the paper is internally consistent with its checked-in artifacts.

App.16 Score-Rank Evaluation Protocol

This appendix states the evaluation protocol as a checklist because many reviewer objections arise from mixing incompatible protocols. The unit of analysis is a problem-conditioned response pool. A response pool contains multiple candidate answers for one benchmark item, each with a correctness label and a scalar score. The selector samples from that pool and returns the highest-scoring candidate. The target metric is the expected correctness of that selected candidate at draw count N.

The first protocol requirement is to keep problem identity fixed. A global pool across all benchmark items is a different experiment. It may estimate how well a score separates easy problems from hard problems, but it does not estimate the behavior of a selector that receives several answers to the same problem. This is why the paper reports the pooled baseline as a failure mode rather than a competing method.

The second requirement is to report the score used for selection, not only the score used for diagnostics. If a system selects by mean log probability but a paper reports a judge AUC, the reported number does not describe the deployed selector. Conversely, if a paper claims judge reranking, the judge score must be attached to every candidate in the response pool or to a clearly specified subset. The theorem is score-agnostic, but the experiment is not score-agnostic unless the score arrays are actually present.

The third requirement is to specify tie handling. In this paper, top-score ties are broken uniformly at random. This convention is mathematically convenient and easy to reproduce. It also avoids giving hidden credit to arbitrary file order. A deterministic tie-breaker can be evaluated, but it must be declared and implemented in the exact-law computation. Otherwise small tie rates can produce confusing mismatches between measured and predicted selection curves.

The fourth requirement is to report the evaluated draw counts. A paper that validates $N \leq 8$ cannot automatically claim behavior at $N = 128$. High-N selection magnifies the top score tail, and the top score tail is exactly where pairwise summaries can fail. The v4 protocol audit therefore reports both moderate draw counts and larger diagnostic draw counts on the 4096-depth slice.

App.17 Exact-Law Implementation Audit

The theorem is simple enough that implementation mistakes are the main risk. The exact-law computation must sort responses by score, group exact ties, and for each tied group compute the probability that the maximum rank among N draws lands in that group. The selected correctness for the group is the fraction of correct responses inside the tied group. Summing the products gives the expected selected correctness.

There are four common bugs. First, using zero-indexed ranks directly in the rank probability term shifts every interval and creates a systematic bias. Second, ignoring ties gives all credit to one arbitrary response in a tied group. Third, pooling problems before applying the law silently changes the experiment. Fourth, estimating the curve by Monte Carlo and then treating simulation noise as theorem error makes the validation weaker than necessary.

The repository guards against these bugs in several ways. The theorem implementation is in `src/theorem.py`. The Du-aligned experiment checks the exact law against empirical measurement over many model/problem/N triples. The v4 protocol script recomputes exact curves from the 4096-depth held-out records and writes the resulting per-record selected accuracies. The $N = 2$ closed form is also a useful unit test: it must match the general score-rank law up to floating-point precision when AUC/kappa uses the same tie convention.

A reviewer who wants to audit the implementation manually can inspect a tiny pool. Suppose four responses have sorted scores $1, 2, 3, 4$ and labels $0, 1, 0, 1$. For $N = 2$, the selected correctness is $(1/4)^2 \cdot 0 + ((2/4)^2 - (1/4)^2) \cdot 1 + ((3/4)^2 - (2/4)^2) \cdot 0 + (1 - (3/4)^2) \cdot 1$. This direct rank calculation is exactly what the code applies at scale, with tied groups replacing single responses when scores are equal.

App.18 High-N Tail Taxonomy

High-N failures are tail failures. The selector does not care about the average score ordering after it has drawn many candidates. It cares about which responses occupy the highest score ranks. This creates several distinct tail regimes.

In a helpful-tail regime, correct responses occupy the top ranks. Increasing N improves selected accuracy because the chance of drawing one of those responses rises. This is the behavior most people implicitly expect from multi-sample inference. It is also the regime where pairwise summaries and high-N accuracy often agree qualitatively.

In a flat-tail regime, correct and incorrect responses are mixed in the upper score ranks. Larger N may help at first and then saturate. Pairwise AUC can look reasonable because the score has some global signal, but extra samples do not produce much selected-accuracy gain once the selector reaches the mixed tail.

In a wrong-tail regime, incorrect responses occupy the highest score ranks. Increasing N can reduce performance even if the score has good pairwise AUC. This is the most important failure for response-pool evaluation because it is precisely the failure that pairwise metrics hide. A small number of confidently wrong answers may have little effect on AUC while dominating max selection.

In a sparse-correct-tail regime, only a few correct responses have extremely high score. AUC can look mediocre because many correct responses are not ranked above many incorrect responses, but high-N selection can be strong if the top tail is correct. This is the mirror image of the wrong-tail failure and is another reason AUC is not a complete high-N statistic.

The score-rank law unifies these cases. It does not ask whether the score is calibrated or whether the score has a high pairwise AUC. It asks how much correctness mass lies in each score-rank interval and how likely max selection is to land there. That is exactly the quantity a multi-sample selector uses.

App.19 Manifest Completion Policy

The repository contains a maxed-out campaign manifest because the strongest future version of the paper would complete a much larger empirical program. The manifest is not treated as complete merely because it exists. Its purpose is to lock targets, command queues, splits, and gates so that future evidence cannot be redefined after seeing results.

The held-out MATH manifest target is 11,500 records: 23 models, 500 problems, and 4096 measured samples per model/problem record. The current v4 paper has 119 records at target depth, which is 1.03% of the manifest. This is why the coverage gate remains missing. The missing gate is not a bug in the audit. It is the audit doing its job.

The live-judge manifest target is also not complete. The completed subset has 135 model/problem pairs and 6,480 judgments. This is enough to show that a live judge score can be attached to a response pool and evaluated by the same law. It is not enough to claim broad judge superiority across all live-runnable models and problems.

The cross-benchmark manifest is similarly larger than the pilot evidence. The completed families are useful for checking whether the exact law survives different graders and task formats. They do not certify six-family manifest-scale generality. The v4 paper therefore uses the phrase “pilot-scale” consistently for these results.

App.20 Result Interpretation By Evidence Tier

The paper separates evidence into tiers. Tier 1 is exact-law validation from completed response-pool artifacts. This is the strongest tier because the law can be evaluated exactly and the measured MAE is near numerical precision. Tier 2 is score-function portability: learned verifiers, live judges, and calibrated scores are different scalar scores on response pools. The law should apply to all of them, and the experiments confirm that it does where the score arrays exist.

Tier 3 is forecasting from finite pilots. This tier is useful but weaker because it estimates an unknown response-pool distribution from a small sample. The expanded-pilot curve improves from small K to larger K, but the curve does not prove that tiny pilots are always enough. The v4 paper reports the curve as sample-complexity evidence, not as a universal stopping rule.

Tier 4 is adaptive allocation. Allocation uses estimated response-pool quantities to decide where additional samples are valuable. The current gains are positive and modest. They support the claim that score-rank estimates are useful for budgeting, not the claim that the proposed allocation policy is globally optimal.

Tier 5 is campaign-scale ambition. This includes full held-out coverage, maxed live-judge coverage, and six-family manifest-scale benchmarks. These are not claimed as completed evidence. They are included so that reviewers can see the roadmap and so that future versions cannot quietly lower the standard.

APP.21 DUPLICATE-RISK FIREWALL

The user-level concern that motivated this v4 pass is real: a stack of papers with similar titles, first theorems, and wrapper contributions can look like template reuse even when the code differs. This paper must therefore have a clear identity when placed beside architecture papers.

The identity here is LLM response-pool evaluation. The mathematical object is a finite scored response pool for one benchmark problem. The empirical objects are MATH response caches, learned verifier scores, live judge scores, pilot cross-benchmark records, and held-out 4096-depth response pools. The failure mode is pairwise evaluation failing to determine high-N selected accuracy. The audit policy is claim gating over completed versus missing evidence.

This is different from a world-model, policy, simulator, retrieval, or planning paper. Those papers need domain mechanisms: dynamics, action selection, planning boundaries, simulator fidelity, retrieval coverage, or control objectives. This paper does not claim those mechanisms. It provides an evaluation law that such papers may use as a diagnostic if they have response or candidate pools with scalar scores and correctness labels.

The manuscript also avoids duplicate cues. The title does not use the repo name. The abstract names pairwise ranking, score-rank moments, LLM response pools, and gated evidence. The introduction contains an explicit boundary against architecture claims. The conclusion says the next step is scale, not a new wrapper. The v4 appendix documents this firewall so a future edit does not accidentally collapse the paper back into a generic template.

APP.22 REVIEWER OBJECTION LEDGER

Objection: the theorem is too simple. Response: the theorem is simple, and the paper does not hide that. The contribution is the exact finite tie-aware law as an evaluation protocol for multi-sample LLM response pools, plus the artifact package showing where pairwise evaluation fails.

Objection: order statistics are known. Response: yes; the paper's novelty is not a claim that maximum ranks are newly discovered. The novelty is applying the exact problem-conditioned rank law to LLM response-pool evaluation and showing that common pairwise summaries are incomplete for high-N selection.

Objection: the full manifest is incomplete. Response: correct. The manuscript states this in the abstract, main text, gate table, gate figure, and appendix. The submitted claim is scoped to completed evidence.

Objection: the live judge subset is too small. Response: it is a scoped subset with 135 model/problem pairs. It supports score portability and a local improvement over log probability on the same pairs; it does not support broad judge superiority.

Objection: the cross-benchmark evidence is pilot scale. Response: yes. The paper uses it to test portability across graders and task families, not to claim manifest-scale generality.

Objection: the held-out slice only covers one model. Response: yes. It is a deep diagnostic slice, not full model coverage. The coverage gate remains missing.

Objection: the adaptive allocation gains are small. Response: yes. The paper reports them as positive but modest. It does not claim optimal allocation.

Objection: AUC is a standard and useful metric. Response: yes. The paper uses AUC where it is complete, at $N = 2$, and then shows why high-N selection needs more information.

APP.23 DATA GOVERNANCE AND NON-FABRICATION

The v4 package follows a strict data governance rule: no result value may be invented in the manuscript. Numbers must come from checked-in artifacts, generated v4 summaries, or command output. If an artifact is absent, the paper must say the corresponding claim is missing rather than filling the gap with optimistic prose.

This rule matters because the repository includes long-running live-collection plans. Some files describe desired targets, some files contain completed measurements, and some files contain partial raw caches. The paper must not confuse these categories. A planned target is not an empirical result. A raw cache below target depth is not a measured held-out record. A missing benchmark family is not covered by analogy to another family.

The v4 protocol audit script helps enforce this rule by producing a compact summary from existing artifacts. The claim audit to be run after the build checks that the v4 ledger exists, the final PDFs exist, stale v2 filenames do not appear in the paper package, and the manuscript discloses missing gates. This does not prove every scientific judgment is correct, but it blocks the most common artifact-level failures.

APP.24 FINAL SUBMISSION CHECKLIST

Before submission, the following checks should pass:

1. The v4 protocol-evidence script runs without network access.
2. The generated macros are included by the paper source.
3. The paper builds to a repository PDF and a visible Desktop PDF named `best-of-n-llm-v4.pdf`.
4. The PDF has at least 25 pages of substantive content.
5. The LaTeX log has no undefined references, citation warnings, fatal errors, emergency stops, or overfull boxes.
6. The v4 claim audit passes.
7. The source map points the Desktop v4 PDF to the local folder and GitHub repo.
8. The GitHub remote head matches the local commit.
9. Stale names such as the old v2 Desktop artifact do not appear in the paper package.
10. Missing gates remain visible rather than being hidden by optimistic wording.

APP.25 WORKED TAIL EXAMPLES

This section gives concrete toy response pools that explain why the exact law is more informative than a pairwise summary. These examples are not additional experiments; they are sanity checks for the reader's intuition.

Consider a pool with ten responses and a score that ranks the only correct response first. The base accuracy is 0.1. At $N = 1$, selected accuracy is 0.1. At $N = 8$, selected accuracy is $1 - (9/10)^8 \approx 0.57$ because the selector succeeds whenever the single correct top-ranked response appears at least once. Pairwise AUC is perfect, but the high-N value is determined by how often the rare top response is sampled.

Now reverse the top tail. Suppose nine responses are correct and one response is incorrect, but the incorrect response has the highest score. The base accuracy is 0.9 and pairwise AUC can still look good if most correct responses outrank most incorrect alternatives in a larger pool. Yet a max selector eventually finds the top-ranked wrong response. In the ten-response toy case, the selected accuracy at $N = 8$ is $(9/10)^8 \approx 0.43$, far below the base rate. This is the wrong-tail failure that matters in high-budget selection.

A third pool has five correct and five incorrect responses arranged in alternating score order. Pairwise AUC is near the middle, and high-N selection depends on the label of the uppermost few ranks. Moving only the top-ranked label can change large-N selected accuracy substantially while barely changing many pairwise comparisons. This is why high-N evaluation is not a smooth function of global pairwise separation alone.

Finally, tied groups require care. If the top score group has four responses, two correct and two incorrect, then the selected correctness conditional on the maximum falling in that tied group is $1/2$, not 1 and not 0. A deterministic file-order tie-breaker could produce a different value, but then the theorem must be changed to match that deterministic protocol. The paper avoids this by using uniform random tie-breaking.

APP.26 BENCHMARK-FAMILY NOTES

MATH is useful for response-pool evaluation because answers are often discrete, grading can be automated with symbolic or exact-answer checks, and large response caches already exist in the repository. The drawback is that MATH is not representative of all language-model tasks. A score that behaves well on MATH can fail on instruction following, code generation, or knowledge tasks. The cross-benchmark pilots address this limitation partially, but not at manifest scale.

GPQA Diamond tests a different competence profile: scientific question answering with multiple-choice structure. The completed pilot shows a gain from $N = 1$ to $N = 48$, but the scale is still 20 tasks. A reviewer should read this as evidence that the exact law can be applied to GPQA-style records, not as a complete GPQA scaling study.

IFEval is valuable because it is instruction-following oriented rather than answer-derivation oriented. Its base accuracy is already high in the pilot, so the headroom for selection gain is smaller. This helps prevent the paper from overfitting its story to difficult math-only settings. Still, a 20-task pilot is only a portability check.

LiveBench selected and LiveCodeBench add more heterogeneous grading behavior. LiveBench selected includes dynamic task families, while LiveCodeBench adds code-oriented evaluation. These pilots are especially important because the exact law should not depend on the semantic type of the task. It only needs scores and correctness labels. The law continues to hold where those arrays exist, but the benchmark breadth claim remains limited by task count and sample depth.

The missing MATH500 and humaneval/MBPP-backed family gates are therefore not cosmetic. They mark real missing evidence. A stronger future paper would run the six-family command queue at the target task and sample scale, then rerun the same v4 protocol audit with updated artifacts.

APP.27 ESTIMATOR SELECTION AUDIT

Estimator selection is a subtle source of accidental overclaiming. If many forecasting estimators are tried on the final held-out set, the best one can look better than it really is. The maxed-out campaign avoids this by using pilot and calibration data for estimator choice, then evaluating locked rows. The v4 paper preserves that distinction and reports the selection source in the held-out tables.

The current locked diagnostic rows use an oracle full-distribution estimator for the completed 4096-depth slice. That estimator is useful for checking the measurement path, exact-law computation, and gate machinery. It is not a deployable finite-pilot estimator because it uses the full measured distribution. The manuscript says this explicitly. Treating the row as a deployable algorithm would be a scientific error.

The expanded-pilot proxy rows are also not final evidence. The existing 256-sample proxy pool does not certify the strongest pilot-forecasting target at all desired K values. The gate report marks one proxy row as WARN and one as INFO. This is valuable because it prevents the paper from saying "sample-efficient forecasting is solved" merely because a narrower curve looks encouraging.

The right interpretation is layered. Exact-law validation is complete for the available full response pools. Forecasting from small pilots is promising but sample-limited. Locked 4096-depth diagnostics show that the measurement system works on 119 records. Full manifest forecasting remains unfinished until coverage is complete.

APP.28 ARTIFACT FAILURE MODES AND MITIGATIONS

Artifact failure mode 1: stale PDF. A repository can have an improved manuscript while the visible Desktop PDF remains an old build. The v4 build script mitigates this by copying the same compiled PDF to `paper/final/best-of-n-llm-v4.pdf` and to the visible Desktop.

Table 11: Claim trace used for the v4 manuscript.

Claim	Evidence source	Boundary
Exact score-rank law has MAE 6.22e-04	`results/du_aligned/du_experiment_results.json`	completed response-pool artifacts only
AUC-only error grows at high N	`results/v4_protocol_evidence/moment_auc_gap_by_n.csv`	not a claim that AUC is useless
Expanded pilots show sample-complexity trend	`results/v4_protocol_evidence/expanded_pilot_by_k_n.csv`	not tiny-pilot sufficiency
Live judge improves over logprob on subset	`results/du_aligned/live_judge_score_summary.json`	135 completed pairs only
Cross-benchmark pilots have exact-law MAE 0	`results/benchmarks/cross_benchmark_summary.csv`	four pilot families only
4096-depth held-out slice is complete for 119 records	`results/maxed_out/heldout_forecasting/measurements/3B`	not full manifest coverage
Adaptive allocation improves over uniform	`results/du_aligned/adaptive_allocation_summary.json`	modest pilot-scale gain

Artifact failure mode 2: stale macros. If result artifacts change but manuscript numbers remain hard-coded, the paper can silently drift away from the evidence. The v4 protocol script writes `v4_results_macros.tex`, and the paper source imports that file.

Artifact failure mode 3: stale claim gates. If the gate report says a gate is missing but the prose says the claim is complete, the paper becomes misleading. The v4 protocol audit checks that missing gates remain disclosed.

Artifact failure mode 4: hidden partial records. Raw caches below target depth can look like progress but should not count as held-out evidence. The gate report counts only records at the required sample depth for the corresponding claim.

Artifact failure mode 5: old title leakage. The repository was previously associated with broader inference-value wording. The v4 paper package should not show the old title as the submission identity. Historical code paths may remain in logs or comments, but the paper, metadata, README, and final audit must use the score-rank LLM evaluation identity.

APP.29 LINE-BY-LINE CLAIM TRACE

The trace table is deliberately redundant with the main text. Redundancy is a feature here: a reviewer should be able to inspect a claim, find the artifact, and see the boundary in one place. This is especially important for a paper that reports both completed evidence and missing ideal gates.

APP.30 WHAT WOULD CHANGE THE CONCLUSION

The central theorem would be falsified by an implementation mismatch, not by a new benchmark result. If the exact law failed on a correctly formed response pool with the stated tie-breaking and sampling protocol, the theorem or code would be wrong. The current evidence makes that unlikely, but the audit still checks it.

The empirical scope would change if future manifest-scale runs contradicted the pilot trends. For example, if six-family manifest-scale benchmarks showed that score-rank evaluation was unstable under real grading noise, the paper would need to narrow its task-generalization language. If live-judge coverage failed to improve over log probability at larger scale, the paper would still support the score-rank law but would drop any positive judge-score claim.

The forecasting story would change if locked finite-pilot estimators failed at larger target depth. That would not invalidate exact response-pool evaluation. It would mean that estimating response-pool tails from small pilots is harder than the current pilot curves suggest. The v4 paper already leaves room for that outcome by treating tiny-pilot sufficiency as unsupported.

The duplicate-risk conclusion would change if the manuscript drifted back to a generic title, generic first theorem, or architecture-free wrapper language without the LLM response-pool evidence. The current v4 version avoids that by making the evaluation target, artifacts, and claim gates specific.

APP.31 SUBMISSION DECISION RATIONALE

The paper is submission-ready only as a scoped evaluation paper. It is not a maxed-out empirical manifesto. The reason to submit the scoped version is that the core law, AUC failure, exact validation, score portability, pilot curves, and claim-gate discipline are already coherent and reproducible. Waiting for the full manifest would improve empirical breadth, but it is not necessary to state the central evaluation lesson honestly.

The strongest version of the abstract must therefore do two things at once. It must lead with the supported contribution: high-N LLM response-pool selection needs conditional score-rank evaluation. It must also disclose the unfinished gates: the held-out 4096-depth evidence is a slice, live-judge coverage is a subset, and cross-benchmark breadth is pilot-scale. This is not weakness laundering. It is how the paper avoids overclaiming while still presenting useful science.

The reviewer most likely to reject the paper will say that the theorem is too simple and the full empirical manifest is incomplete. The answer is that the paper is not selling theorem difficulty or completed manifest scale. It is selling a precise evaluation protocol, backed by enough artifacts to show that the protocol matters and enough gate discipline to show where it does not yet reach.

APP.32 RESPONSE-POOL REPORTING STANDARD

The main text argues that a response-pool selector is not characterized by one pairwise number. This appendix turns that argument into a reporting standard that can be checked by a reviewer or by a future benchmark maintainer. The standard is intentionally concrete because loose reporting is one reason that multi-sample language-model claims are hard to compare. A paper that reports a "best of many" number without the response-pool law leaves open whether the gain came from better scoring, easier problems, duplicate responses, lucky pool composition, or a hidden aggregation choice.

For each benchmark family, we require five objects. First, the paper must name the unit being conditioned on. In this manuscript the unit is a model-problem response pool with a fixed score function and a binary correctness label. Second, it must report the sampling protocol. The law in the paper uses draws with replacement from the measured empirical pool, with uniform random tie-breaking among tied selected responses. Third, it must report the score rank distribution, not just an AUC. Fourth, it must separate exact problem-conditional quantities from pooled or model-level summaries. Fifth, it must state which records are missing from the intended manifest and how those missing records affect the claim.

The v4 artifact package follows this contract. The exact law is validated on 35,964 conditional triples from the Du-aligned response pools. The held-out 4096-depth slice is reported as 119 completed records out of an intended 11,500 manifest, not as a completed manifest. The cross-benchmark evidence is explicitly four pilot families, and the live-judge evidence is explicitly 135 model-problem pairs with 6,480 judgments. These numbers are not decorative metadata; they are part of the claim. If any one of them is removed, a reviewer can no longer tell whether a stated response-pool gain is an exact finite fact, a pilot trend, or an unfinished target.

This standard also explains why the paper does not use the old generic "Best-of-N" framing as its scientific identity. The actual contribution is a language-model evaluation protocol for score-rank response pools. The protocol has a formal core, but the submission is judged by whether the measured response pools, gates, and missing-manifest boundaries support the language model claims. A sibling

Table 12: Minimum reporting contract for high-N LLM response-pool claims.

Item	Required disclosure	Why it matters
Conditioning unit	Model, benchmark, problem, score function, label rule, and response count.	Prevents pooled summaries from hiding problem-level failures.
Sampling rule	Whether selected responses are drawn with replacement, without replacement, or from a deployment-specific generator.	The finite law and tail probability depend on this rule.
Tie rule	How equal scores are handled before selecting a response.	Ties are common for verifier scores and rubric scores, and they change high-N behavior.
Rank evidence	At least enough conditional rank moments to recover the claimed high-N accuracy, or an exact empirical computation.	AUC only identifies the two-draw case and can be misleading at larger N.
Manifest status	Completed records, excluded records, unavailable endpoints, and failed gates.	Makes empirical scope auditable instead of rhetorical.

Table 13: Aggregation choices and their reviewer interpretation.

Aggregation	Acceptable use	Failure mode
Problem-conditional exact law	Primary scientific claim; computes selector accuracy inside each response pool.	Requires the response pool and tie rule to be fixed and documented.
Model mean over problems	Useful summary after conditional quantities are computed.	Can hide a bimodal benchmark with many trivial and many impossible records.
Benchmark-family mean	Useful for comparing benchmark families at pilot scale.	Can overstate generality when families differ in grading noise or response format.
Pooled score distribution	Diagnostic negative control.	Invalid as a primary estimate when problem difficulty and score calibration vary.
Live-judge aggregate	Useful scoped comparison when all judged pairs are disclosed.	Can become a provider-quality claim if missing endpoints or judged subsets are not reported.

paper about simulators, planners, or world models could share some order-statistic algebra; it could not reuse this reporting standard without changing the objects, scores, labels, and failure modes.

APP.33 PER-PROBLEM VERSUS PER-MODEL AGGREGATION

The safest way to read a response-pool experiment is from the conditional problem outward. Let $A_{m,p,N}$ denote the exact expected accuracy of a selector at sample count N for model m and problem p. The central empirical object is this conditional quantity, or its score-rank equivalent. Only after those conditional quantities are computed should the paper average over problems, models, or benchmark families. Reversing the order can turn a heterogeneous benchmark into a single smooth-looking curve whose uncertainty has no clear interpretation.

The pooled baseline in the main text is included for precisely this reason. It aggregates response scores before respecting the problem-conditioned law and therefore obtains a much larger error than the exact conditional computation. This is not an implementation footnote. In multi-sample LLM evaluation, easy problems can have many correct high-score responses, while hard problems can have no correct responses or only rare correct responses. Pooling first gives the easy records more leverage over the score distribution and then applies that distorted distribution to records where it does not belong.

The practical recommendation is to keep three ledgers. The first ledger stores record-level exact quantities and is the only ledger allowed to support exact claims. The second ledger stores model-level summaries and confidence or bootstrap intervals after record-level computation. The third ledger stores scope flags: missing records, excluded endpoints, grader failures, degenerate pools, and calibration-only artifacts. The v4 protocol-evidence script implements this separation by writing conditional held-out records, benchmark family deltas, gate statuses, and macros as separate files. The build then imports the macros rather than retyping values in the manuscript.

This aggregation discipline is especially important for high N. At $N = 2$, pairwise AUC has a direct interpretation because the selector compares two responses. At $N = 48$ or $N = 256$, the selected response is governed by the upper tail of the score-rank distribution. A model with a modest mean improvement can still fail on records where the score places rare correct answers below many attractive incorrect answers. Conversely, a model with noisy scores can look weak in pairwise terms but still place a small number of correct answers in the top tail. The per-problem law is what distinguishes these cases.

APP.34 REVIEWER REPRODUCTION STRESS TESTS

The manuscript should be easy to attack locally. A reviewer does not need to rerun the entire response-generation campaign to test whether the main claims are coherent. The following stress tests are deliberately small enough to run from checked-in artifacts, but each one targets a real failure mode.

1. Recompute the exact law on the Du-aligned triples and confirm that the mean absolute error remains near 6.22e-04. This catches theorem-code mismatches, rank-order mistakes, and tie-handling drift.

2. Recompute the AUC-only approximation at $N = 48$ and compare it with the exact law. The expected failure is about 0.0821 MAE, not zero. If the AUC baseline suddenly looks exact at high N, the test is probably no longer measuring the intended high-N selector.

3. Recompute the pooled baseline. It should remain worse than the conditional computation, with the v4 pooled MAE around 0.122. This catches accidental aggregation before conditioning.

4. Sample the 4096-depth held-out slice and verify the boundary statement: the slice has 119 completed records, median correctness rate 0.348, and only 1.03% of the intended manifest. The correct conclusion is scoped evidence, not manifest completion.

5. Recreate the claim-gate figure and table. The result should preserve 8 passed gates and 4 missing gates. A submission draft that deletes the missing gates is less reliable, not more polished.

6. Compare live-judge and log-probability selection on the judged subset. The live-judge value at $N = 48$ is 0.561 versus 0.496 for log probability, but the claim remains a subset result.

These tests are intentionally adversarial. They assume that an implementation may silently change a sign convention, a rank convention, a tie rule, a record filter, or a file path. They also assume that an author may be tempted to phrase a pilot as a completed manifest. The audit does not merely check that figures compile; it checks that the manuscript still says the thing the artifacts justify.

The most important stress test is semantic rather than numerical. Place this paper next to a paper about world models, planners, diffusion policies, or trajectory transformers. If the first theorem, first page, and experiment claims look interchangeable, the manuscript has failed even if every number is correct. The current paper avoids that failure by making LLM response pools, score functions, judge subsets, problem-conditioned aggregation, and manifest-gated evidence visible throughout the paper. A reviewer should be able to describe the paper as a response-pool evaluation paper without seeing the repository name.

APP.35 SCORE GAMING AND DEPLOYMENT THREAT MODEL

The score-rank law is descriptive: it tells us what a fixed score function does on a fixed empirical response pool. It does not certify that the score function is causally robust, immune to gaming, or safe for deployment. This distinction matters because high-N selection can amplify both good ranking signal and bad proxy incentives. A weak score that occasionally gives extreme values to incorrect but fluent responses can look tolerable at $N = 2$ and then become dangerous at $N = 48$ or $N = 256$.

The threat model has four parts. First, score hacking: responses may optimize the surface form preferred by a judge, verifier, or rubric without improving truth. Second, duplicate collapse: a generator may produce many near-identical responses, reducing the effective response-pool diversity while preserving a large nominal N. Third, grader mismatch: a score calibrated for one benchmark family may reward the wrong behavior on another family. Fourth, selection-induced distribution shift: a high-N selector repeatedly exposes only the tail responses, so downstream users see a distribution different from the generator's ordinary sample distribution.

The experiments in this paper do not solve those deployment risks. They make them easier to measure. A response-pool report that includes rank moments, tie rates, degenerate pools, exact conditional curves, and missing gates gives reviewers a way to ask whether the top tail is trustworthy. The live-judge subset is useful in this sense: it tests one richer score source against a log-probability baseline, but it is not broad enough to declare judge scores universally safe. The cross-benchmark pilot is also useful: it checks whether the exact-law machinery survives different grading surfaces, but it is not a guarantee of all-domain calibration.

For deployment, the conservative rule is to treat a high-N gain as a claim about a measured response pool, not about an unconstrained future generator. Before using such a selector in a new setting, one should rerun the conditional law on fresh response pools, inspect failure cases from the selected top tail, and compare at least one negative control score. The paper's exact law makes that rerun cheap once responses are measured; the expensive part remains the collection of representative responses and trustworthy labels.

APP.36 RELEASE AND ARCHIVE CONTRACT

The final manuscript is only as reproducible as its archive. The build process therefore treats generated PDFs, figures, macros, and claim reports as a submission package rather than as disposable local state. The v4 contract is simple. The repository stores the source, scripts, checked-in cached evidence, and final PDF. The visible Desktop stores only the final versioned PDF for this paper. The source map points a fresh agent from the Desktop PDF to the local folder and GitHub repository. The GitHub repository name, local folder name, and Desktop PDF stem agree.

Table 14: Archive contract used for the v4 submission package.

Artifact	Required state	Audit reason
Desktop PDF	`best-of-n-llm-v4.pdf` only for the current final artifact.	Fresh sessions can find the final paper without stale drafts.
Repository PDF	`paper/final/best-of-n-llm-v4.pdf`.	The pushed repository contains the same submission artifact as the Desktop.
Source package	Versioned v4 source ZIP when packaging is requested.	Submission upload can be recreated from repository state.
Cached evidence	`results/v4_protocol_evidence/summary.json` plus CSV ledgers and figures.	Numeric manuscript claims can be traced without new API calls.
Claim audit	A script-level audit that checks page count, stale names, gates, and hashes.	Prevents cosmetic rewrites from silently weakening the evidence boundary.

This contract is partly administrative, but it has scientific consequences. If the Desktop contains an older draft while the repository contains a v4 source tree, a fresh agent or reviewer can easily inspect the wrong paper. If the repository PDF and Desktop PDF have different hashes, there is no single submission artifact. If stale documentation still names an old generic theorem identity as the current title, duplicate-wrapper risk reappears even though the manuscript itself was rewritten. The archive contract prevents these failures from becoming invisible.

The contract also constrains future improvements. A later v4 may add the full held-out manifest, additional benchmark families, or broader live-judge coverage. If so, it should not edit v4 numbers in place. It should generate a new evidence directory, new macros, a new final PDF name, and a new audit note. Versioned evidence is less glamorous than a single polished number, but it is what lets reviewers distinguish completed experiments from future ambition.

APP.37 LARGE-LANGUAGE-MODEL USAGE DISCLOSURE

Codex, an OpenAI language model accessed through the local coding-agent interface, materially assisted with this manuscript package. Its role was to organize the paper from verified repository artifacts, draft and edit LaTeX text, create tables and packaging scripts, run local build and hygiene checks, and help phrase the claim boundaries. The human authors remain responsible for all scientific claims, formal statements, artifact selection, and final submission decisions. No result value was invented by the model; numeric claims were traced to repository artifacts listed in Appendix App.4.